

Problem-Solution Fit

Date	28 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Students struggling with specific topics during self-study - Educators, TAs & academic support teams - Online course learners seeking quick help - Trainers and corporate learning teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited access to one-on-one instructor time - Varying device/internet quality - Time constraints during self-study - Limited awareness of own emotional state while learning	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: Generic chatbots, forum posts, static FAQs Pros: - Easily accessible - Free or low-cost Cons: - No emotional awareness - Generic, one-size-fits-all replies - Slow human response on forums
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty recognizing why they feel stuck (Very Frequent) - Frustration goes unnoticed until it affects performance (Frequent) - Generic help doesn't match emotional state (Frequent) - Delayed responses from instructors (Occasional)	9. PROBLEM ROOT / CAUSE [RC] - No real-time way to detect learner emotion from text - Manual sentiment interpretation is slow and inconsistent - Lack of personalized, emotion-aware guidance - Support resources are not tailored to how the learner feels	7. BEHAVIOR + ITS INTENSITY [BE] - Searches online for quick explanations (High) - Re-reads material repeatedly when confused (Medium) - Avoids asking for help until frustrated (Medium)
3. TRIGGERS TO ACT [TR] - Getting stuck on a concept or assignment - Feeling bored, confused, or frustrated while studying - Preparing for an exam or deadline	10. YOUR SOLUTION [SL] The Emotion Detection & Learning Support Engine uses BiLSTM and BERT models with Gemini AI to detect a learner's emotional state from free text and respond with tailored guidance. - Classifies emotion (Bored, Confident, Confused, Curious, Frustrated) from student text - Generates personalized tips, next steps, and encouragement - Compares BiLSTM vs BERT for reliable, transparent predictions - Logs interactions and visualizes emotion trends on a dashboard	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Course forums - Search engines - Messaging apps with TAs OFFLINE - Classrooms - Study groups - Office hours

4. EMOTIONS BEFORE / AFTER [EM]

Before: Confused, frustrated, bored,
uncertain

After: Confident, curious, motivated,
reassured